

Intelligent Model Selection Architecture

How Model Decisions are Made from Prompts | Air-Gapped Agentic AI Workbench

Core Routing Principle: In an on-premise, air-gapped system, using a single generic LLM for every task causes high latency, GPU VRAM waste, and context bottlenecks. The system uses a **deterministic 5-stage selection pipeline** that inspects prompt semantics, keywords, file attachments, and active VRAM budgets to select the optimal model in **< 2 milliseconds**.

1. The 5-Stage Model Selection Pipeline

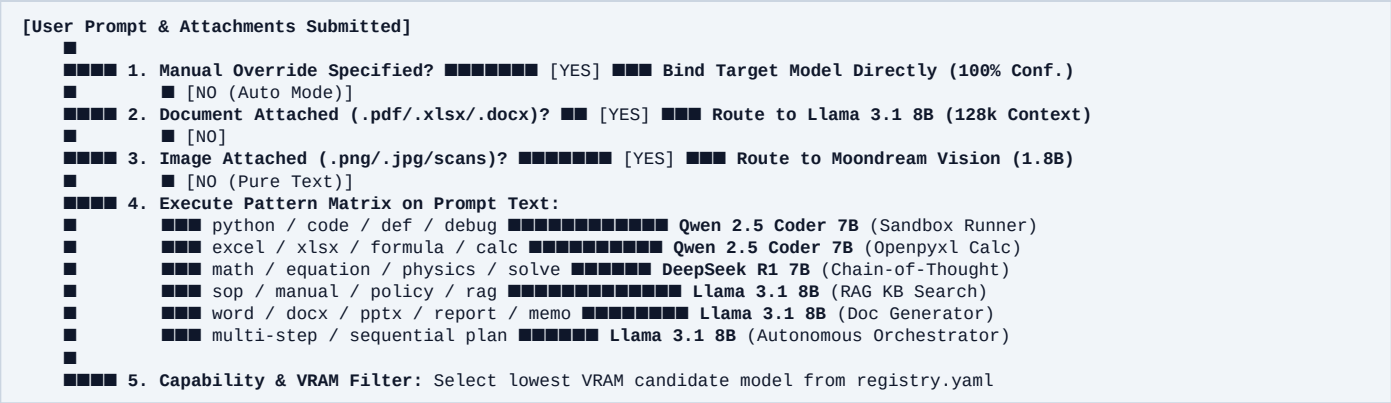
Incoming prompts and tasks pass through a hierarchical priority ladder before model invocation:

- **Stage 1 — Explicit User Override:** If the user manually chooses a model (e.g. qwen2.5-coder:7b or deepseek-r1:7b), the router immediately bypasses auto-classification with 100% confidence.
- **Stage 2 — Multimodal & Document Attachment Detection:** Non-image files (.pdf, .docx, .xlsx, .csv, .txt) auto-route to **Llama 3.1 8B** (128k context window). Image files (.png, .jpg, scanned PDFs) auto-route to **Moondream Vision** for visual inspection.
- **Stage 3 — Regex Pattern Matrix & Intent Classification:** For pure text prompts in Auto mode, the TaskClassifier runs regex pattern matching against 6 capability domains (Coding, Spreadsheet formulas, Document drafting, RAG SOPs, Mathematics, and Summarization).
- **Stage 4 — Capability Mapping against Dynamic Registry:** The classified intent is mapped to registered model capability profiles defined in registry.yaml.
- **Stage 5 — VRAM Budgeting & Hardware Optimization:** If multiple models match, the router selects the candidate with the **lowest VRAM footprint** to prevent GPU Out-Of-Memory (OOM) errors.

2. Prompt Trigger Matrix & Assigned Models

Prompt / Attachment Triggers	Detected Intent	Assigned Model	Ctx	VRAM	Primary Role & Capability
python, def, script, debug, algorithm, regex, execute code	Code Generation	Qwen 2.5 Coder 7B (qwen2.5-coder:7b)	32k	4.7 GB	Python code synthesis, script debugging, automated sandbox verification.
excel, spreadsheet, xlsx, csv, formula, sumif, vlookup, pivot	Spreadsheet / Calc	Qwen 2.5 Coder 7B (qwen2.5-coder:7b)	32k	4.7 GB	Openpyxl workbook manipulation, financial models, formula calculation.
word, docx, pptx, presentation, approval note, summary, memo	Document Drafting	Llama 3.1 8B (llama3.1:8b)	128k	4.9 GB	Structured deliverable generation, professional formatting, executive memos.
sop, manual, guideline, policy, regulation, clause, standard	RAG Policy Search	Llama 3.1 8B (llama3.1:8b)	128k	4.9 GB	Vector knowledge base retrieval, semantic grounding, strict policy adherence.
solve, equation, physics, calculus, derivation, proof, math	Math & Deep Reasoning	DeepSeek R1 7B (deepseek-r1:7b)	32k	4.7 GB	Step-by-step chain-of-thought derivations, engineering and physics computations.
image, drawing, p&id, diagram, scan, blueprint, photo, OCR	Vision & OCR	Moondream 1.8B (moondream)	4k	2.0 GB	Visual diagram OCR, scanned equipment schematic reading, visual QA.
first... then... finally, step 1, step 2, multi-tool task	Multi-Step Orchestration	Llama 3.1 8B (llama3.1:8b)	128k	4.9 GB	Chains vision extraction, spreadsheet analysis, and report generation.

3. Visual Decision Logic Flowchart



4. Dynamic Model Registry & Hot-Swapping

The model selection architecture is decoupled from code via a dynamic registry (registry.yaml). This provides three key operational benefits:

- **Zero-Downtime Hot Registration:** New quantized models (e.g. deepseek-coder-v2:16b or custom fine-tuned models) can be registered dynamically via REST API without restarting the orchestrator daemon.
- **Automatic Model Tag Normalization:** If a user requests a model tag without a version suffix (e.g., qwen2.5-coder vs qwen2.5-coder:7b), the selector resolves the tag against the local Ollama disk library automatically.
- **Fault-Tolerant Fallback Cascading:** If an assigned specialized model is unavailable or fails, the execution graph automatically cascades to the resident general reasoning model (llama3.1:8b) to ensure task completion.

5. Summary of Model Selection Benefits

Selection Metric	System Implementation	Operational Benefit in Air-Gapped Environment
Routing Latency	< 2.0 milliseconds (Regex + In-Memory Table)	Zero token cost and zero pre-call LLM latency overhead.
Context Optimization	128k window for docs, 32k for code/math	Prevents token truncation on large financial spreadsheets or PDFs.
VRAM Efficiency	Least-VRAM-First prioritization (2.0GB – 4.9GB)	Enables complete agentic execution on a single workstation GPU.
Security & Air-Gap	100% on-device Ollama execution (localhost)	Zero external data egress; complete compliance with air-gapped security.